Devansh Shah

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SUMMARY

AI Engineer specializing in LLMs, RAG systems, semantic search, and machine learning. M.S. in Artificial Intelligence (3.93 GPA), ACM SIGCSE 2026 publication, and internship experience delivering production AI solutions across research and industry.

SKILLS

Programming Languages: Python, R, SQL, C++
AI & LLMS: GenAI, RAG & Adaptive RAG Pipelines, LlamaIndex, AI Agents, LLM Finetuning & Hallucination Mitigation, LangChain, Transformers, Dense Vector Retrieval, Cross-Encoder Re-ranking, Hugging Face, Llama, Qwen, GPT & Gemini models.
ML & Data Science: Recommender Systems, Predictive Modeling, Classification, Regression, Time-Series Forecasting, Feature Engineering, Model Evaluation, Exploratory Data Analysis (EDA), Machine Learning, Deep Learning, Statistics, PyTorch, TensorFlow, Scikit-learn, Pandas, Numpy, Data Cleaning, Data Visualization, Data Preprocessing, Web Scraping & Crawling.
Databases: MongoDB, Vector Databases, PostgreSQL(pgvector), Supabase.
Backend: FastAPI, Flask, Django, Streamlit, Docker, Kubernetes, CI/CD Pipelines, Git, DSA
Other: SLURM Batch Scripting, HPC Cluster Deployment, Parallelism (ThreadPoolExecutor), Power BI, VS Code, Render, Kaggle

EDUCATION

The University of Texas at San Antonio **Aug 2024 – May 2026**
Masters in Artificial Intelligence (with Computer Science track) GPA: 3.93
G.E.C., Gandhinagar (GTU) **Jun 2019 – Jun 2023**
Bachelor of Engineering, Computer

PUBLICATIONS

Tian, N., **Shah, D.**, Sanusi, I., & Martin, F. (2026). **Ice Cream Express: A Game-Based Tool to Teach AI Pathfinding to K-12 Learners**. Accepted to SIGCSE Virtual 2026, The ACM Virtual Global Computing Education Conference.

EXPERIENCE

University of Texas at San Antonio *Researcher, San Antonio, TX* **Sep 2026 – Present**
- Drive end-to-end NLP, LLM and Generative AI research by identifying open problems in literature, engineering Python-based models, developing RAG and semantic retrieval pipelines, running data experiments, and co-authoring academic publications.

San Antonio African American Community Archive and Museum *Intern, San Antonio, TX* **Jan 2026 – Apr 2026**
- Developed an LLM-powered chatbot and semantic search tools leveraging NLP, RAG, embeddings, SQL databases, and vector databases to enhance natural-language access to museum archives and historical content.
- Collaborated with non-technical staff on generative AI, workflow automation, and AI training to improve efficiency and accessibility.

Quacito LLC *Trainee Data Science Engineer (Intern), San Antonio, TX* **Jun 2025 – Aug 2025**
- Developed LLM-based sentiment and service-identification agents integrating RAG pipelines, vector databases (MongoDB), semantic retrieval, and OpenAI models for reviews, feedback, and work-order data.
- Designed and optimized prompts, contributing to end-to-end AI solution development in a fast-paced startup environment

Downtown Technologies Pvt. Ltd. *Data Science Intern, Ahmedabad, India* **Jan 2023 – May 2023**
- Built an Elasticsearch-based search system for 1.2M+ Indian brand products, securing 99.5% accuracy (30% improvement); handled crawling, preprocessing, categorization, ranking, and UI design using JavaScript.(GitHub: shyamsar30/flask-app-elasticsearch)
- Automated email verification, price recommendations, and image processing; used Celery for job scheduling and large-scale crawling (1.2M+ products), reducing manual workload; trained categorization models using Random Forest and Logistic Regression on AWS EC2, stored datasets in S3, and improved categorization accuracy by 2%, recommendations by 5%, and user interaction by 10%.

University of Texas at San Antonio *Teaching/Lab Assistant* **Sep 2024 – May 2026**

PROJECTS

Ongoing Project: AI-Powered Video Teaser Generator: Building an AI-powered video understanding pipeline that generates teaser clips from long-form YouTube videos utilizing OCR, Whisper transcription, and LLM-based contextual analysis. Implemented the Phase 1 backend harnessing Python, FastAPIs, and REST APIs for video ingestion, subtitle extraction with Whisper fallback, OCR text detection, metadata processing, and multimodal context preparation.
Adaptive Multi-Hop RAG: Built an end-to-end Adaptive RAG pipeline for multi-hop QA on MuSiQue (19,938 questions) using Llama 3.3 70B, BGE-M3 retrieval, and Cross-Encoder reranking; engineered NLP/LLM workflow with question decomposition, semantic retrieval, retry-based refinement and answer extraction, achieving 0.450 EM (+81 % vs published oracle baseline). Scaled large-batch experiments on UTSA HPC with SLURM and 16-worker ThreadPoolExecutor parallelism; deployed an interactive Streamlit UI with Docker and GitHub Actions CI/CD for automated production deployment.(GitHub: DevanshShah2002/Adaptive_Rag)
LLM Fine-Tuning Pipeline: Phi-3.5-Mini-Instruct (PEFT + SLURM): Fine-tuned Phi-3.5-Mini-Instruct with a two-stage instruction tuning pipeline (Alpaca-Cleaned + custom teacher-generated data) using PEFT/TRL on a SLURM-managed HPC cluster, demonstrating end-to-end LLM training, evaluation, and local deployment. (GitHub: DevanshShah2002/NLP_Assignment_3)
LLM Debate & Multi-Agent Reasoning: Built a multi-agent debate pipeline (Llama 3.1, Qwen) for StrategyQA, gaining 74.5% accuracy leveraging adaptive stopping, specialized role prompting, and a judge model.(GitHub: DevanshShah2002/NLP_Assignment_2)

LEADERSHIP & ADDITIONAL INFORMATION

International Student Advisory Council Member, UTSA; Volunteer Bird rescuer, SAJAG(1000+ birds rescued Jan 2021 - Jun 2024)
Certifications: Google Data Analytics Professional Certificate; Pre-MBA Statistics, Indian Institute of Management Ahmedabad